

systemd

systemd is the init system and service manager on most modern Linux distros. `systemctl` controls units (services); `journalctl` reads their logs. Most commands need `sudo` to change state.

Control Services

Command	Action
<code>systemctl start nginx</code>	start a service now
<code>systemctl stop nginx</code>	stop it now
<code>systemctl restart nginx</code>	stop then start
<code>systemctl reload nginx</code>	reload config without a full restart
<code>systemctl enable nginx</code>	start automatically at boot
<code>systemctl disable nginx</code>	don't start at boot
<code>systemctl enable --now nginx</code>	enable at boot and start immediately

Check Status

Command	Shows
<code>systemctl status nginx</code>	state, recent logs, and PID
<code>systemctl is-active nginx</code>	<code>active</code> / <code>inactive</code> (scriptable)
<code>systemctl is-enabled nginx</code>	whether it starts at boot
<code>systemctl --failed</code>	every unit that failed to start
<code>systemctl list-units --type=service</code>	all loaded services
<code>systemctl list-unit-files</code>	all installed units + their state

Read Logs (journalctl)

Command	Shows
<code>journalctl -u nginx</code>	all logs for one service
<code>journalctl -u nginx -f</code>	follow (tail) live
<code>journalctl -b</code>	logs since the last boot
<code>journalctl -e</code>	jump to the end
<code>journalctl --since '1 hour ago'</code>	time-filtered logs
<code>journalctl -p err</code>	errors and worse only
<code>journalctl -k</code>	kernel messages (like dmesg)
<code>journalctl --disk-usage</code>	how much space the journal uses

Unit Files

Command	Does
<code>systemctl cat nginx</code>	print the unit file and any overrides
<code>systemctl edit nginx</code>	create a drop-in override (safe)
<code>systemctl edit --full nginx</code>	edit a full copy of the unit
<code>systemctl daemon-reload</code>	reload after editing unit files
<code>systemctl show nginx</code>	dump every property

Always daemon-reload after editing. systemd caches unit files. Change one and forget `sudo systemctl daemon-reload` and it keeps running the old definition — a classic "why isn't my change working" trap.

System State

Command	Does
<code>systemctl reboot</code>	restart the machine
<code>systemctl poweroff</code>	shut down
<code>systemctl suspend</code>	sleep
<code>systemctl get-default</code>	current boot target (e.g. graphical)
<code>systemd-analyze blame</code>	what slowed down the last boot